

Indirect Prompt Injection Propagation in Agentic AI Pipelines: A Structural Analysis of Attack Surface and Exploitation Pathways

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Abstract

When an AI system retrieves information from outside sources to answer a question or complete a task, it implicitly trusts what it reads. This trust creates a vulnerability: an attacker who plants hidden instructions inside a document, webpage, or database record can cause the AI to act on those instructions rather than the user's original request without the user ever knowing. The AI does not malfunction. It does exactly what it is designed to do. The problem is that it cannot distinguish legitimate content from adversarial content embedded within it.

This vulnerability becomes significantly more dangerous in multi-step AI systems, where individual agents pass outputs to one another, each treating the previous agent's result as trusted input. An instruction injected at one stage can travel through the entire system and trigger real-world consequences far removed from where the attack entered.

This work examines how hidden instructions propagate through three representative pipeline structures: a sequential step-by-step chain, a parallel arrangement where multiple agents work simultaneously, and a retrieval-based system where the AI looks up sources before responding. Using an AI research assistant as the primary demonstration, the methodology employs a controlled set of sources — a mix of benign documents and one adversarial document containing a hidden instruction — and observes whether that instruction alters the system's final output. Results are analyzed by propagation depth, output integrity change, and whether the system produces any detectable signal of compromise.

The primary contribution is a structured taxonomy table linking pipeline design properties to exploitation risk and defender visibility, giving developers a concrete reference for identifying and addressing injection vulnerabilities before deployment.

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